

SWASTIK CHOWDHURY

ssamadda@alumni.cmu.edu | 412-844-1548 | [LinkedIn](#) | [Github](#)

EDUCATION

Carnegie Mellon University

August 2024 - August 2025

Master of Information Systems

Coursework: Machine Learning, AI Engineering (Teaching Assistant), Database Systems, Distributed Systems, DevSecOps, Data Focused Python (Teaching Assistant), Object-Oriented Programming with Java

MAKAUT

June 2015 - June 2019

Bachelor of Technology

Coursework: Computer Programming, Data Structures & C, Object-Oriented Programming, Database Management

WORK EXPERIENCE

National Transfer Network, *Data Science*

October 2025 - Present

- Built a RAG system over internal company documentation and unstructured data, giving students natural-language access to institutional knowledge.
- Implemented semantic retrieval with ChromaDB and built an evaluation pipeline to compare embedding models and reduce hallucinations, using the results to select the best-performing configuration.

Vertisage Technologies, *Software Engineer - Data*

April 2023 - March 2024

- Developed a financial invoice data extraction pipeline using LlamaIndex, Pydantic, and AWS Textract OCR to transform semi-structured PDF into schema-validated Parquet tables.
- Led migration of **80+** legacy Redshift MPP stored procedures to Apache Spark (Python) ETL on EMR with Airflow orchestration, cutting annual infrastructure cost by **\$1M+** and reducing job runtime by **40%** while maintaining full data correctness.
- Implemented a Debezium-based Change Data Capture pipeline writing to Iceberg tables on S3, delivering sub-**5-minute** updates to fraud analytics datasets consumed by the Risk team.
- Built event-driven monitoring and automated alerting with SNS/SQS/Lambda, reducing **MTTD** and improving SLA adherence from **95% to 99.5%** for user behavioral telemetry datasets.

Tata Consultancy Services, *Software Engineer - Data*

July 2019 - April 2023

- Optimized Apache Spark (Python) ETL by diagnosing data skew via Spark UI stage metrics and applying salting and broadcast joins cutting runtime by **40%** and cloud compute cost by **15%** across production pipelines.
- Owned primary on-call rotation for **25+** production Spark/Airflow pipelines serving **4 downstream teams**, authoring runbooks and automating recurring failure modes to cut repeat incidents by **60%** and reduce **MTTR from 45 min to 15 min**.
- Built Apache Spark (Python) ETL pipelines with Airflow to ingest **5+ GB/day** of user-interaction JSON events into S3, partitioning by date and event type to cut Athena scan volumes and query costs by **40%** and drop analyst query latency from minutes to seconds.
- Designed dimensional data marts using **dbt** incremental models and **Snowflake** clustering strategies, improving reporting performance and reducing warehouse compute costs by **20%**

ACADEMIC PROJECTS

Recommendation Systems (SCS CMU)

(1M users, 27K movies, <500 ms latency, 99% availability)

- Built real-time data pipelines ingesting from Kafka and orchestrating data using Airflow into MySQL.
- Containerized Flask services with Docker, deployed them to Kubernetes
- Setup Jenkins CI/CD, and ensured observability with Grafana and Prometheus.
- Measuring traceability with MLflow and also checked for data drift using KL divergence.

SKILLS

Programming: Python, Java, Scala, C++, SQL

Databases: MySQL, Redshift, Snowflake, Iceberg, Delta Lake, Presto/Trino

Data Engineering: Apache Spark, Hadoop, Kafka, Airflow, Flink, dbt, Databricks

Cloud & DevOps: AWS, Docker, Kubernetes, Terraform, Jenkins, Prometheus, Grafana, Linux (Bash), Postman, Git